SentinelVision AI

Volume 08

DevOps and Deployment

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> **SentinelVision AI — Enterprise Documentation Suite**

> Status legend: Implemented | Planned | Future

1. Environments

Environment	Purpose	Status
Local	Developer workstation	Implemented
CI	Automated validation	Implemented
Staging	Pre-production	Planned
Production	Customer-facing	Planned

2. CI/CD — GitHub Actions

Current pipeline jobs:

- backend: migrate + pytest
- edge: pytest
- frontend: lint + build
- prd/docs: documentation build

Planned gates: coverage thresholds, security scan, container scan, staging deploy.

3. Docker

Image	Base	Purpose
backend	python:3.12-slim	Django API
frontend	node:20-alpine + nginx	Dashboard
edge	python:3.12-slim	Edge worker

4. Docker Compose

Services: db, backend, frontend, edge with healthchecks and persistent postgres volume.

5. Kubernetes (Planned)

- Helm chart with values per environment
- Separate namespaces for staging/production
- Secrets via external secrets operator

6. Nginx

- Serve frontend static assets
- Proxy `/api/` to backend
- Planned: TLS termination, rate limits, WAF integration

7. SSL/TLS

- Local dev: HTTP
- Production: TLS 1.2+ with auto-renewed certificates

8. Redis & Celery (Planned)

Worker deployment for notifications, reports, retention cleanup.

9. Monitoring

Component	Tool	Status
Metrics	Prometheus	Planned
Dashboards	Grafana	Planned
Logs	ELK	Planned
Uptime	Health checks	Partial

10. Backup & DR

Asset	RPO	RTO	Method
PostgreSQL	15 min	1 hr	Managed backups + PITR
Object storage	1 hr	2 hr	Cross-region replication
Config/secrets	24 hr	4 hr	Git + vault restore

11. Security Hardening

- Non-root containers
- Secret rotation policy
- Network policies in Kubernetes
- Dependabot and bandit in CI (Planned)

12. Runbooks

Scenario	Action
API unhealthy	Check DB connectivity, restart pods

Scenario	Action
Edge offline	Verify outbound HTTPS, API key, RTSP source
Ingest backlog	Scale API replicas, inspect edge queue
DB failover	Promote replica, update connection string